2025 MIT Science Bowl High School Invitational

Round 8

TOSS-UP

1) CHEMISTRY – *Multiple Choice* Gideon performs an organic reaction and isolates a sample of the product, which is pure up to stereoisomerism (read: *stereo-eye-SAW-murr-ism*). He tests the sample and finds that it is optically inactive. Which of the following could not be the composition of the product?

W) Mixture of enantiomers (read: *en-ANN-tee-oh-murrs*)
X) Mixture of diastereomers (read: *dye-uh-STEH-ree-oh-murr*)
Y) Single pure enantiomer
Z) Single pure diastereomer

ANSWER: Y) SINGLE PURE ENANTIOMER

BONUS

1) CHEMISTRY – *Short Answer* Rank the following three conformations of n-butane by increasing energy: 1) Gauche; 2) Eclipsed; 3) Anti.

ANSWER: 3, 1, 2

TOSS-UP

2) PHYSICS – *Multiple Choice* An object is placed 3 centimeters from a convex lens with a focal length of 5 centimeters. Which of the following describes the type and size of the image compared to the object, respectively?

W) Real and larger
X) Real and smaller
Y) Virtual and larger
Z) Virtual and smaller

ANSWER: Y) VIRTUAL AND LARGER

BONUS

2) PHYSICS – *Short Answer* An uniform magnetic field in the x-direction has magnitude 2 teslas. In newtons, what is the z-component of the force it exerts on a 1-coulomb point charge with velocity vector $\hat{\imath} + 2\hat{\jmath}$ (read: *i hat plus 2 j hat*) meters per second?

ANSWER: -4

TOSS-UP

3) EARTH AND SPACE – *Multiple Choice* At which of the following points along a stream profile would you expect to see the most sediment particles moving via saltation?

W) Headwaters
X) Floodplain
Y) Delta
Z) Mouth

ANSWER: W) HEADWATERS

BONUS

3) EARTH AND SPACE – *Short Answer* Identify all of the following three phenomena that are associated with an El Niño (read: *L-NEEN-yo*) event: 1) Weakened Walker circulation; 2) Fewer Atlantic hurricanes; 3) Westerly wind bursts.

ANSWER: ALL

TOSS-UP

4) MATH – *Short Answer* The sum of the first n terms of an infinite geometric sequence with first three terms $1/4$, $1/8$, and $1/16$ is $63/128$. What is the value of n?

ANSWER: 6

BONUS

4) MATH – *Short Answer* A bag contains 4 red balls and 5 orange balls. Boheng draws two balls from the bag, without replacement. Given that at least one of the drawn balls is red, what is the probability that both drawn balls are red?

ANSWER: 3/13

TOSS-UP

5) ENERGY – *Multiple Choice* Scientists at the Picower (read: *PICK-ow-ur*) Institute have been studying certain neurotransmitters that result in excitotoxicity. Which of the following neurotransmitters are they likely studying?

W) Glutamate
X) GABA
Y) Substance P
Z) Serotonin

ANSWER: W) GLUTAMATE

BONUS

5) ENERGY – *Short Answer* Scientists from the Laub Lab developed an anti-bacteriophage defense system. Their system uses an enzyme to attack bacteriophage mRNA by covalently attaching an ADP molecule linked to what five-carbon carbohydrate?

ANSWER: RIBOSE

TOSS-UP

6) BIOLOGY – *Short Answer* While the coleoptile (read: *koh-lee-OP-tile*) covers the plumule (read: *PLOOM-yool*), the coleorhiza (read: *koh-lee-uh-RYE-zuh*) is responsible for covering and protecting what structure?

ANSWER: RADICLE

BONUS

6) BIOLOGY – *Short Answer* Instead of petals, some plants like euphorbs (read: *you-forbes*) use what decorative modified leaves associated with flowers to attract pollinators?

ANSWER: BRACTS

TOSS-UP

7) PHYSICS – *Short Answer* Thin spherical shells A and B of the same thickness are constructed from sheets of the same uniformly dense material. If the diameter of A is twice the diameter of B, what is the ratio of the moment of inertia of A to the moment of inertia of B about axes passing through their respective centers?

ANSWER: 16

BONUS

7) PHYSICS – *Short Answer* David has an isolated parallel plate capacitor with a plate separation of 1 centimeter that stores an energy of 1 joule. He then separates the two plates to twice their original separation. How much work, in Joules, did he do?

ANSWER: 1

TOSS-UP

8) BIOLOGY – *Short Answer* Density-dependent inhibition in cancer cells is analogous to what process in bacteria, responsible for regulating biofilm formation?

ANSWER: QUOROM SENSING

BONUS

8) BIOLOGY – *Short Answer* Boheng's chrysanthemum seeds are about to germinate, but midterms are looming. If Boheng wants to delay germination until they get their free time back, identify all of the following three treatments they could apply to the seeds: 1) Abscisic (read: *ab-SISS-ik*) acid; 2) Far-red light; 3) White light.

ANSWER: 1 AND 2

TOSS-UP

9) CHEMISTRY – *Short Answer* If the first half-life of a zeroth order reaction with one reactant is 10 seconds, how long in seconds will the reactant take to be completely consumed?

ANSWER: 20

BONUS

9) CHEMISTRY – *Short Answer* Identify all of the following three gas-phase molecules that are linear: 1) Beryllium difluoride; 2) Xenon difluoride; 3) Silicon dioxide.

ANSWER: ALL

TOSS-UP

10) EARTH AND SPACE – *Multiple Choice* Which of the following features would most likely hinder thunderstorm development?

W) High surface humidity
X) Temperature inversion in lower atmosphere
Y) Strong vertical wind shear
Z) Converging surface winds

ANSWER: X) TEMPERATURE INVERSION IN LOWER ATMOSPHERE

BONUS

10) EARTH AND SPACE – *Multiple Choice* A walrus and a carpenter were walking down a beach on January 1 at 12am, when they noticed that the water had reached the very top of the intertidal zone. Given that this beach has mixed semidiurnal tides, when will sea level next reach this maximum height?

W) January 1 at 12pm
X) January 7 at 12pm
Y) January 8 at 12am
Z) January 15 at 12am

ANSWER: Z) JANUARY 15 AT 12AM

TOSS-UP

11) MATH – *Short Answer* A perfect number is a positive integer equal to the sum of its proper divisors. What is the smallest positive integer *n* such that the number of positive divisors of *n* is a perfect number?

ANSWER: 12

BONUS

11) MATH – *Short Answer* What is the smallest three digit palindrome divisible by 3 and 4?

ANSWER: 252

TOSS-UP

12) ENERGY – *Short Answer* The Fedorenko Lab is studying the development of the brain's language production functions. What part of the brain is responsible for production of language?

ANSWER: BROCA'S AREA

BONUS

12) ENERGY – *Multiple Choice* The Weiss Group studied samples from Ryugu, a C-type asteroid. Which of the following statements about Ryugu is true?

W) Ryugu lies in the outer edge of the asteroid belt
X) Ryugu reflects the majority of incident sunlight
Y) Ryugu is less dense than ice
Z) Ryugu crystals exhibit Widmanstatten (read: *VID-mun-shtah-tun*) patterns

ANSWER: W) RYUGU LIES IN THE OUTER EDGE OF THE ASTEROID BELT

TOSS-UP

13) BIOLOGY – *Multiple Choice* Which of the following treatments is most likely to yield minimal bacterial growth in the Ames test for mutagenicity?

W) Addition of Ethidium Bromide
X) Addition of Ascorbic Acid
Y) Exposure to UV radiation
Z) Addition of Reactive oxygen species

ANSWER: X) ADDITION OF ASCORBIC ACID

BONUS

13) BIOLOGY – *Short Answer* Kat finds a fish that produces extensive amounts of TMAO in its tissue. Identify all of the following three statements that would likely be true about this fish: 1) It has a lateral line; 2) It has jaws; 3) It is ovoviviparous.

ANSWER: ALL

TOSS-UP

14) PHYSICS – *Short Answer* The period of oscillations of an LC circuit is proportional to capacitance raised to what power?

ANSWER: 1/2

BONUS

14) PHYSICS – *Short Answer* Tim the beaver stands 40 centimeters from a concave mirror with a focal length of 10 centimeters. What is the absolute value of the image distance, in centimeters and rounded to two significant figures?

ANSWER: 13

TOSS-UP

15) ENERGY – *Multiple Choice* The Van Voorhis Group developed super-resolution techniques for electronic spectra. Which of the following techniques could best be used to resolve the constituent frequencies of a signal?

W) Principal component analysis
X) Singular value decomposition
Y) Fourier transform
Z) Cholesky decomposition

ANSWER: Y) FOURIER TRANSFORM

BONUS

15) ENERGY – *Short Answer* Researchers in MIT's Research Laboratory of Electronics are studying superconductivity in twisted bilayer graphenes. Superconductivity in these materials can be confirmed by the presence of what bosonic bound states predicted by BCS theory?

ANSWER: COOPER PAIRS

TOSS-UP

16) CHEMISTRY – *Multiple Choice* Which of the following couplings is responsible for generating multiplets in NMR spectroscopy?

W) Protons of adjacent nuclei
X) Electrons of adjacent atoms
Y) Vibrations of adjacent atoms
Z) Excitations of adjacent fluorophores

ANSWER: W) PROTONS OF ADJACENT NUCLEI

BONUS

16) CHEMISTRY – *Short Answer* Identify all of the following three elements that form a stable one to one binary compound with nitrogen: 1) Copper; 2) Antimony; 3) Strontium.

ANSWER: 2 ONLY

TOSS-UP

17) MATH – *Short Answer* A rhombus has sides of length 5 and diagonals with integer side lengths. What is the area of the rhombus?

ANSWER: 24

BONUS

17) MATH – *Short Answer* The least common multiple of 120 and a positive integer is 2520. What is the least possible value of the positive integer?

ANSWER: 63

TOSS-UP

18) EARTH AND SPACE – *Short Answer* A red giant has a hundred times the radius and half the temperature of our Sun. What is its luminosity, in solar luminosities?

ANSWER: 625

BONUS

18) EARTH AND SPACE – *Short Answer* Identify which of the following three parameters that, when increased, would increase a molecular cloud's Jeans length: 1) Temperature; 2) Density; 3) Mean molecular mass.

ANSWER: 1 ONLY

TOSS-UP

19) PHYSICS – *Multiple Choice* Which of the following subatomic particles does not interact via the weak force?

W) Higgs boson
X) Neutrino
Y) Up quark
Z) Gluon

ANSWER: Z) GLUON

BONUS

19) PHYSICS – *Short Answer* An ideal massless spring is resting on a frictionless table with zero rest length and a spring constant of 24 newtons per meter. One end of the spring is fixed, while the free end is attached to a point mass of 6 kilograms. Penelope gives the mass an initial velocity of 5 meters per second perpendicular to the spring. In meters, what is the radius of the point mass's circular motion?

ANSWER: 2.5 (ACCEPT: 5/2 (READ: *FIVE OVER TWO*))

TOSS-UP

20) ENERGY – *Short Answer* MIT engineers developed a technique based on dynamic light scattering, or DLS, to estimate nanoparticle size distributions in powders. In DLS, incident light is elastically scattered by particles smaller than its wavelength. What type of scattering is this an example of?

ANSWER: RAYLEIGH SCATTERING

BONUS

20) ENERGY – *Short Answer* The Quantum Engineering Group measured the correlation between observations made by quantum sensors. Identify all of the following three numbers that could be a measured Pearson correlation: 1) -0.5; 2) 0.5; 3) 1.5.

ANSWER: 1 AND 2

TOSS-UP

21) BIOLOGY – *Multiple Choice* Dr. Katherine is doing karyotypes of her patients. Which of the following patients' karyotypes would have a normal number of chromosomes?

W) A patient with Klinefelter and cri du chat (read: *cree-doo-shaa*) syndrome
X) A patient with Klinefelter and Prader-du Willi syndrome
Y) A patient with Turner and Down syndrome
Z) A patient with Turner and Angelmann syndrome

ANSWER: Y) A PATIENT WITH TURNER AND DOWN SYNDROME

BONUS

21) BIOLOGY – *Short Answer* Identify all of the following three places where one could find portal systems in the human body: 1) Median eminence of infundibulum (read: *in-fun-DIB-yuh-lum*); 2) Hepatic vein; 3) Circle of willis in the brain.

ANSWER: 1 AND 2

TOSS-UP

22) EARTH AND SPACE – *Multiple Choice* Which of the following is not a significant energy source for Jupiter?

W) Residual heat from formation
X) Solar heating
Y) Kelvin-Helmholtz contraction
Z) Tidal heating

ANSWER: Z) TIDAL HEATING

BONUS

22) EARTH AND SPACE – *Short Answer* A star of apparent magnitude 6 is observed to have parallax angle 1 milli-arcsecond. What is its absolute magnitude?

ANSWER: −4

TOSS-UP

23) CHEMISTRY – *Multiple Choice* Mingwen is measuring the absorbance of a copper (II) sulfate solution to determine its concentration. Which of the following colors of light would be best suited for this experiment?

W) Red
X) Green
Y) Blue
Z) White

ANSWER: W) RED

BONUS

23) CHEMISTRY – *Multiple Choice* Edward roasts 100 grams of calcium sulfate at high temperature and finds that the mass of the resulting sample is 79 grams. Given that the molar mass of anhydrous calcium sulfate is 136 grams per mole, which of the following is the most likely number of waters of hydration in the original sample?

W) 1
X) 2
Y) 3
Z) 4

ANSWER: X) 2

TOSS-UP

24) MATH – *Multiple Choice* Which of the following is closest to the area under the curve $y = 1 + (\sin(x))^{2025}$ (read: *y equals 1 plus open parentheses, sine of x, close parentheses to the power of 2025*) from $x = 0$ to $\pi/2$?

W) 0.8
X) 1.6
Y) 1.9
Z) 3.1

ANSWER: X) 1.6

BONUS

24) MATH – *Short Answer* Consider a tetrahedron of side length 1. A number is uniformly randomly selected between 0 and the tetrahedron's height; then, a cross-section is drawn at that height above the base. What is the cross-section's expected area?

ANSWER: $\sqrt{3}/12$

TOSS-UP

25) BIOLOGY – *Multiple Choice* Dr. Pujan is treating his patient a Sam with a unique disease where noncoding LINE elements are hyperactive, causing mutagenesis. Knowing this, which of the following noncoding RNAs would be MOST effective in treating Sam?

W) miRNA (read: *em-eye-RNA*)
X) siRNA (read: *ess-eye-RNA*)
Y) piRNA (read: *pee-eye-RNA*)
Z) snRNA (read: *ess-en-RNA*)

ANSWER: Y) PIRNA (READ: *PEE-EYE-RNA*)

BONUS

25) BIOLOGY – *Short Answer* A wobble base pair is a pairing between two nucleotides in RNA molecules that does not follow Watson-Crick base pair rules. In tRNA, what nucleobase is present that contributes to three of the four main wobble base pairs?

ANSWER: HYPOXANTHINE (ACCEPT: INOSINE)